

NanoGT Match Report: Severe combined immunodeficiency due to adenosine deaminase deficiency

Disease: Severe combined immunodeficiency due to adenosine deaminase deficiency (ORPHA:277)

Primary gene: ADA

Gene CDS: 1092 bp

Inheritance: Autosomal recessive

Target tissues: hematopoietic

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Strimvelis	LV	7.8/10	[GREEN] High	approved
2	Libmeldy	LV	6.4/10	[YELLOW] Medium	approved
3	Skysona	LV	6.3/10	[YELLOW] Medium	approved
4	AVR-RD-01	LV	6.0/10	[YELLOW] Medium	phase1/2
5	BMN 307	AAV5	5.8/10	[YELLOW] Medium	phase2

Match #1: Strimvelis

Precedent disease: ADA-SCID

Vector: LV

Tissue target: hematopoietic

Composite score: 7.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector

Dimension	Score	Max	What it measures
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	7.83	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 1092bp / cargo 8000bp (14% utilized)
- Vector tropism plus precedent target match: hematopoietic
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: immune_hematopoietic
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated in lentiviral ex vivo programs (ADA-SCID, beta-thalassaemia)
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders

Match #2: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	6.44	10.0	Raw sum / 18×10

Rationale

- Gene CDS 1092bp / cargo 8000bp (14% utilized)
- Vector tropism plus precedent target match: hematopoietic
- Protein class mismatch
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (immune_hematopoietic vs leukodystrophy)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated in lentiviral ex vivo programs (ADA-SCID, beta-thalassaemia)
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders

Match #3: Skysona

Precedent disease: Cerebral adrenoleukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	6.28	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 1092bp / cargo 8000bp (14% utilized)
- Vector tropism plus precedent target match: hematopoietic
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Different pathway (immune_hematopoietic vs peroxisomal)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity

- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated in lentiviral ex vivo programs (ADA-SCID, beta-thalassaemia)
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders

Match #4: AVR-RD-01

Precedent disease: Fabry disease

Vector: LV

Tissue target: hematopoietic

Composite score: 6.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	6.00	10.0	Raw sum / 18×10

Rationale

- Gene CDS 1092bp / cargo 8000bp (14% utilized)
- Vector tropism plus precedent target match: hematopoietic
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Different pathway (immune_hematopoietic vs lysosomal_storage)
- Approval status: phase1/2
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated in lentiviral ex vivo programs (ADA-SCID, beta-thalassaemia)
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders

Match #5: BMN 307

Precedent disease: Phenylketonuria

Vector: AAV5

Tissue target: liver

Composite score: 5.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.30	1.0	Immunological protection of target tissue
Promoter availability	0.70	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	5.83	10.0	Raw sum / 18×10

Rationale

- Gene CDS 1092bp / cargo 4700bp (23% utilized)
- No tissue overlap (disease: ['hematopoietic'], vector: ['liver', 'lung', 'CNS'])
- Both intracellular proteins
- Inheritance match (Autosomal recessive <-> AR)
- Different pathway (immune_hematopoietic vs amino_acid_metabolism)
- Approval status: phase2
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: low privilege — immune cells reside here; robust conditioning and monitoring required
- Promoter availability: EFS, PGK, SFFV — validated in lentiviral ex vivo programs (ADA-SCID, beta-thalassaemia)
- Route of administration: Ex vivo HSC delivery — technically complex but highly precise; gold standard for blood disorders